

Module 1 assignment

- Draw an ER diagram for the following application from the manufacturing industry and map the ER- model formed to relational schema.

Each supplier has a unique name.

More than one supplier can be located in the same city.

Each part has a unique part number.

Each part has a colour.

A supplier can supply more than one part.

A part can be supplied by more than one supplier.

A supplier can supply a fixed quantity of each part.

2. A university keeps track of student_name, current_add, ph_no, dob

(i) Some user applications need to refer to the city, state and zip_code of the students address.

(ii) Each department described by a name, dept_code, office_phn.

Both name and code has unique value for each department.

(iii) Each course has a course_name, description, course_no, no_of_semester_hrs. The value of course_no is unique.

(iv) Each section has a faculty, semester_yr, course and sec_no. The sec_no is unique.

Draw the E-R diagram for the above schema. Specify key attributes. If any unspecified requirement is there make appropriate assumption and complete it.

3. Drawing of ER model of university database application considering the constraints –

- A university has many departments.
- Each department has multiple instructors (one person is HOD). Here the HOD refers to the head of department.
- An instructor belongs to only one department.
- Each department offers multiple courses, each subject is taught by a single instructor.
- A student may enroll for many courses offered by different departments.

4. Draw the system structure of DBMS. Explain the functionalities of the following components of DBMS: DDL Compiler, Query Compiler, precompiler, runtime database processor.

5. Convert the following ER-diagram into equivalent relational model.

6. Suppose you are given the following requirements for a simple database for the National Hockey League (NHL):

- i) The NHL has many teams,
- ii) Each team has a unique name, a city, a coach, a captain, and a set of players,

- iii) Each player belongs to only one team,
 - iv) Each player has a unique ID, a name, a position (such as left-wing or goalie), a skill level, and a set of injury records,
 - v) A team captain is also a player,
 - vi) A game is played between two teams referred to as host_team and guest_team
 - vii) The date of the game and the score in the game is recorded.
- Construct a clean and concise ER diagram for the NHL database to include all the mentioned requirements.

7. The Flight Database stores details about an airline's fleet, flights, and seat bookings. Consider the following requirements list:

- The airline has one or more airplanes.
- An airplane has a model number, a unique registration number, and the capacity to take one or more passengers.
- An airplane flight has a unique flight number, a departure airport, a destination airport, departure date and time, and arrival date and time.
- Each flight is carried out by a single airplane.
- A passenger has a name, a surname, and a unique email address.
- A passenger can book a seat on a flight.

Construct an ER diagram to meet all these constraints and requirements.

Medium Question

- Define the following terms: Domain constraint, Entity integrity and Referential integrity.
- A weak entity has existence dependency on strong entity. However, every existence dependency doesn't imply a weak entity set. State [True/False] and justify your answer with an example.
- Explain the relational model terminologies, Relation, Tuple, Degree, Cardinality and Constraint.
- Explain the roles and responsibilities of DBA in an organization.
- What is a recursive relationship in ER Model? Explain it with a suitable example?